Hierarchical Semantic Propagation for Object Detection in Remote Sensing Imagery

Chunyan Xu, Chengzheng Li, Zhen Cui, Tong Zhang, and Jian Yang

***Abstract*— Object detection in remote sensing imagery is a critical yet challenging task in the field of computer vision due to the bird's-eye-view perspective. Although existing object detection approaches in remote sensing imagery have achieved great advances through the utilization of deep features or rotation proposals, but they give insufficient consideration to multilevel semantic information and its propagation for guiding the learning process. Accordingly, in this article, we propose a hierarchical semantic propagation (HSP) framework to boost object detection performance in remote sensing imagery, which is better able to propagate hierarchical semantic information among different components in a unified network. Given a remote sensing image as input, the HSP framework can detect instances of semantic objects belonging to certain categories in an end-to-end way. First, the multiscale representation is captured by a basic feature pyramid network, which can hierarchically combine spatial attention details and the global semantic structure in order to learn more discriminative visual features. Second, the soft-segmentation prediction is used as an auxiliary objective in the intermediate layer of our HSP; its output instance-aware semantic information can be propagated to suppress noisy background information and thereby guide the proposal generation in the region proposal network. By further propagating this hierarchical semantic information into the region of interest module, we can then predict the object category information and the corresponding horizontal and oriented bounding boxes. Comprehensive evaluations on three benchmark data sets demonstrate the superiority of our HSP to the existing state-of-the-art methods for object detection in remote sensing imagery.**

***Index Terms*— Hierarchical semantic propagation (HSP), object detection, remote sensing imagery.**

NOMENCLATURE

HSP	Hierarchical semantic propagation.
CNN	Convolutional neural network.
FPN	Feature pyramid network.
RoI	Region of interest.
RPN	Region proposal network.
RRPN	Rotation RPN.
HBB	Horizontal bounding box.
OBB	Oriented bounding box.
SPP	Spatial pyramid pooling.
ASPP	Atrous SPP.
IP	Image pooling.
PSP	Pyramid scene parsing.
GCN	Global convolutional network.
R^2CNN	Rotational region CNN.
R-CNN	Region-based CNN.
fc layer	Fully connected layer.
mAP	Mean Average Precision.

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I. INTRODUCTION

AS REMOTE sensing technology has rapidly evolved, high-quality satellite and aerial images have become the primary data source for various applications, including natural disaster management, surface migration analysis, meteorological observation, traffic monitoring, and urban road construction. Meanwhile, object detection, particularly in remote sensing imagery, has gradually become a significant research topic in the field of computer vision. However, due to the variation in terms of the posture and altitude of the aircraft when collecting images, objects in remote sensing imagery are always characterized by several specific characteristics distinct from those of objects in natural scenes; these include the arbitrariness of the arrangement, the disunity of the ground sample distances, the high complexity of the background, etc. Due to the restrictions imposed by these complex factors, the task of object detection in remote sensing imagery also faces a number of significant challenges that are not directly addressed by general object detection techniques.

Due to their powerful capabilities of deep feature representation, CNNs have become an essential part of numerous computer vision tasks in recent years, including image classification [1]–[4], semantic segmentation [5]–[7], and object detection [8]. Some previous CNN-based object detection approaches have also facilitated great advances in natural scenes, such as the detection algorithms belonging to the R-CNN family [9], [10], single shot detector (SSD) [11], and YOLO [12]–[14]. For example, R-CNN, the classic object detection algorithm [8], first generates category-independent region proposals, adopts a CNN model for extracting a fixed-length feature vector from each region, and then predicts the corresponding category with support vector machines (SVMs). Fast R-CNN [9] and faster R-CNN [10] methods then integrate deep feature extraction, the generation of region proposals, the prediction of an object class, and location information in an end-to-end network. A real-time object detection system utilizing YOLO [12]–[14] has also been proposed to address the object detection task from a regression perspective

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with a single-stage algorithm. Moreover, the previous approaches [5], [7], [15], [16] have taken the detection and segmentation networks into joint training, which can improve the performance of both tasks. Besides, a few detection algorithms [17], [18] have utilized the weak box-wise segmentation results to enhance feature representation based on some classic detectors, which demonstrates some extra gains of semantic information for the detection task. Although the above object detection approaches have achieved very promising performance on natural images, the challenges still remain when utilizing them directly to solve the object detection problem in remote sensing imagery, due to the unique characteristics of input data.

Recently, researchers have attempted to utilize existing modern CNN-based detectors to design more effective approaches [19]–[23] to detect objects in remote sensing imagery. On the one hand, the improvement of deep features can be used to address the aerial object detection problem [24]–[26], especially when the scales of objects in remote sensing images are quite inconsistent. For example, based on VGG-16, an end-to-end single CNN [24] has been proposed to detect vehicles with arbitrary orientation in aerial imagery. Ding *et al.* [22], use the FPN [27] as backbone, then learn an RoI transformer aimed at solving the oriented object detection problem. On the other hand, several other aerial object detection methods [19], [21], [26], [28] adopt RRPNs [29], which can generate rotation region proposals regressed from skew anchors that can better predict oriented objects in aerial images. Although methods utilizing skew anchors can obtain better coverage of oriented objects in aerial scenes, they also obviously lead to an increased computational burden due to the generation of dozens or even hundreds of anchors per pixel position. Furthermore, R^2CNN [30], designed to detect arbitrary-oriented objects, has been used to regress these rotation bounding boxes from horizontal region proposals in remote sensing imagery [31], [32], which can to some extent reduce the amount of computation required. When performing crop operations (e.g., RoIAlign [5] or RoIPool [9]), however, these R^2CNN-based detection methods would bring in too much background noise or too many other adjacent objects, especially in aerial scenes where objects are characterized by arbitrary orientations, complex backgrounds, and dense arrangements. Moreover, these existing aerial detection methods are mainly concerned with the arbitrary orientations of objects but fail to consider how multilevel and instance-aware semantic information might be utilized and then propagated throughout the learning process in order to further improve object detection performance, especially in remote sensing imagery.

In this article, we propose a novel HSP framework[1] to deal with the object detection problem in remote sensing imagery. Our proposed HSP not only employs instance-aware segmentation prediction but also propagates the hierarchical semantic information throughout the network learning process. More specifically, we adopt the basic FPN to hierarchically capture the spatial attention details and global semantic structure, which can better facilitate the learning/optimization of multiscale discriminative visual features. In the middle layer of the HSP framework, we design a soft-segmentation prediction module with an auxiliary objective function; its instance-aware semantic output can then be propagated to suppress noisy background information and enable a focus on the foreground information in the generation of horizontal region proposals (i.e., the RPN module). By further propagating this hierarchical semantic information into the semantic-fused RoI (S-RoI) module, we can learn more distinguished object representation for predicting the object category information and location results for both HBBs and OBBs. Our proposed HSP achieves 80.42% mAP and 78.01% mAP in HBB and OBB prediction tasks of the DOTA test set, which represent better performance than existing state-of-the-art algorithms [32]–[34]. We also obtain state-of-the-art results on two other public data sets, namely the UCAS_AOD and NWPU VHR-10 data sets. The main contributions of this article can be summarized as follows.

1) We propose an HSP framework through jointly learning multiscale visual representation and instance-aware semantic information in a unified network, in order to boost the performance of object detection in remote sensing imagery.
2) We design a semantic-guided RPN, which utilizes the instance-aware semantic information to suppress the noisy background information and better generate horizontal region proposals.
3) We design an S-RoI module to integrate multilevel semantic features and appearance features as the enhanced representation for the classification and localization of HBBs/OBBs.

II. Related Work

A. Object Detection in Natural Scenes

Much research has been devoted to object detection in natural scenes recently [8]–[14], [35], including one-stage and two-stage detectors. Two-stage object detection networks typically propose a set of candidate boxes for the input image sample and then process them with classic classifiers. In the R-CNN Girshick *et al.* [8], first introduced CNN into detection tasks for obtaining a remarkable performance. Faster R-CNN [10] brought in the RPN, which integrates the detection task into an end-to-end framework on the basis of SPP [36] and Fast R-CNN [9]. Lin *et al.* [27] proposed FPNs to fuse multiscale features in order to improve the detection performance. Some single-stage detectors have also been proposed, including YOLO [12]–[14] and SSD [11]. For instance, RetinaNet [37] achieved better performance than two-stage detectors while retaining the efficiency of single-stage ones. However, these methods are not specifically designed for the object detection problem in remote sensing imagery due to the bird's-eye-view perspective.

[1]This is an extensive version of our article originally presented at the CVPR19 workshop, named "Learning Object-Wise Semantic Representation for Detection in Remote Sensing Imagery." Moreover, in the DOAI2019 challenge, we ("pca_lab") achieved the first and second place on the HBBs and OBBs tasks of the DOTA-v1.5 data set, respectively. More information can be found at https://captain-whu.github.io/DOAI2019/results.html

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Several previous works [5], [7], [15], [16] have attempted to utilize semantic information when dealing with the detection task. For example, Dai *et al.* [16] presented an effective multitask cascade framework to simultaneously conduct instance-aware semantic segmentation and detection results; Mask R-CNN [5] used a parallel branch to predict the segmentation mask in the Faster R-CNN framework [10]; Hybrid task cascade method [7] utilized the mask information flow and spatial context features to enhance both detection and segmentation predictions; the others [17], [18] used weak box-wise segmentation outputs to enhance feature representations. The main differences between our HSP and these above methods lie in two folds: 1) ours uses semantic segmentation as a constraint stream for detection, while these methods usually employ semantic segmentation as concatenated channel features to enhance representation and 2) ours only need weak mask semantic information generated from skew bounding boxes, whereas these methods either need groundtruth of segmentation [5], [7] or employ HBBs [17], [18].

B. Object Detection in Aerial Scenes

Some generic detection algorithms [20], [23]–[25], [33], [38], [39] in aerial scenes have utilized the horizontal mode of bounding boxes. The classic R-CNN frameworks [8], [10] were adopted in [24], [25], [33], [38], [40], and [41] to learn rotation-invariant detectors in aerial images. For instance, Zhang *et al.* [42] proposed an improved R-CNN method for boosting the detection result of small and gathering ships, which employs multiresolution convolutional features and performs the RoI pooling operation on a larger feature map in the RPN module. Deformable Convolutional Networks [43] has been widely applied to detection tasks in aerial images [44], [45] and achieved promising results in several aerial data sets. YOLT [20] combined two detection models [12], which are trained on different aerial image scales to achieve better detection results for objects of disparate sizes. Liu *et al.* [39] proposed to use feature maps from layers of different depths and an oriented response dilated convolution strategy to improve multiclass aerial object detection based on YOLOv2 [13]. Moreover, the recurrent detection with activated semantics (RDAS) approach [46] aimed to boost aerial object detection by adopting semantic segmentation-aware CNN features in order to activate the detection feature maps. Yan *et al.* [23] analyzed the different roles played by different IoU thresholds in the detection network and accordingly propose an IoU-adaptive deformable R-CNN framework for multiclass aerial object detection.

Inspired by some modern scene text detectors [29], [30], various arbitrary-oriented aerial detection algorithms have also emerged. Following RRPN [29], many researchers attempted to incorporate rotation region proposals into the detection framework to regress OBBs. For example, based on the RRPN framework, DRBox [28] used a special multiangle prior rotation anchor strategy to handle the aerial objects' arbitrary orientation. Zhang *et al.* [26] proposed a novel rotated RoIPool technique, which was combined with RRPN to facilitate oriented ship detection. R-DFPN [21] utilized a dense FPN with multiscale RoIAlign to detect ships in different scenes based on rotation anchors [29]. In image cascade network (ICN) Azimi *et al.* [19] proposed a joint image cascade and FPN to fuse multiscale features from multiscale images, which can regress both horizontal and OBBs with the help of rotation region proposals [29]. Li *et al.* [47] proposed R^3Net, which used three rotation-specific subnetworks to predict multioriented vehicles in aerial images and videos. Furthermore, by extending R-DFPN [21] with adaptive RoIAlign and multitask learning, Yang *et al.* [32] proposed a rotational region ship detection framework that can predict the prow direction, HBBs and OBBs. R^2CNN++ [32] was designed to complete rotational object detection via a special anchor sampling strategy that is based on both channel and pixel attention subnetworks. Compared with these above existing detection methods, our HSP considers the utilization and propagation of hierarchical semantic information (such as multiscale representations of RPN and instance-aware segmentation predictions) in the optimized process of the detection network.

III. Proposed HSP

A. Basic Formulation

As illustrated in Fig. 1, the proposed HSP framework consists of the basic FPN, the instance-aware soft segmentation module, the semantic-guided RPN, and the S-RoI module. Given a remote sensing image $X \in \mathbb{R}^{W\times H\times 3}$ as input, we first extract the multiscale feature representation using a basic FPN, which can make full use of low-level details and high-level semantic features with a classic top-down and bottom-up network structure. This process can be formulated as $R = f_{\text{FPN}}(X)$, where f_{FPN} denotes the feature extraction function from FPN, while R refers to the multiscale representation to be propagated to the following modules. By feeding the multiscale representation R into the soft segmentation module, we can obtain the instance-aware segmentation result S, such that the auxiliary objective information can be roughly estimated from the OBBs of the training set in the learning process. The instance-aware soft segmentation module can be formulated as

$$S = f_{\text{seg}}(R) \tag{1}$$

where f_{seg} refers to the function of obtaining instance-aware segmentation outputs by implementing a sequence of convolutional layers on the multiscale features R in practice. The predicted instance-aware semantic information S and the multiscale representation R can be then aggregated into the semantic-guided RPN, where it can not only work to suppress the noisy background information but also enable more attention to be paid to these meaningful object regions. The generation of region proposals in semantic-guided RPN module can be formulated as follows:

$$P = f_{\text{RPN}}(R, S) \tag{2}$$

where P denotes the predicted region proposals with the semantic-guided RPN module, i.e., f_{RPN}. The S-RoI module f_{RoI} can consider hierarchical semantic features simultaneously: these include the multiscale representation of FPN R,

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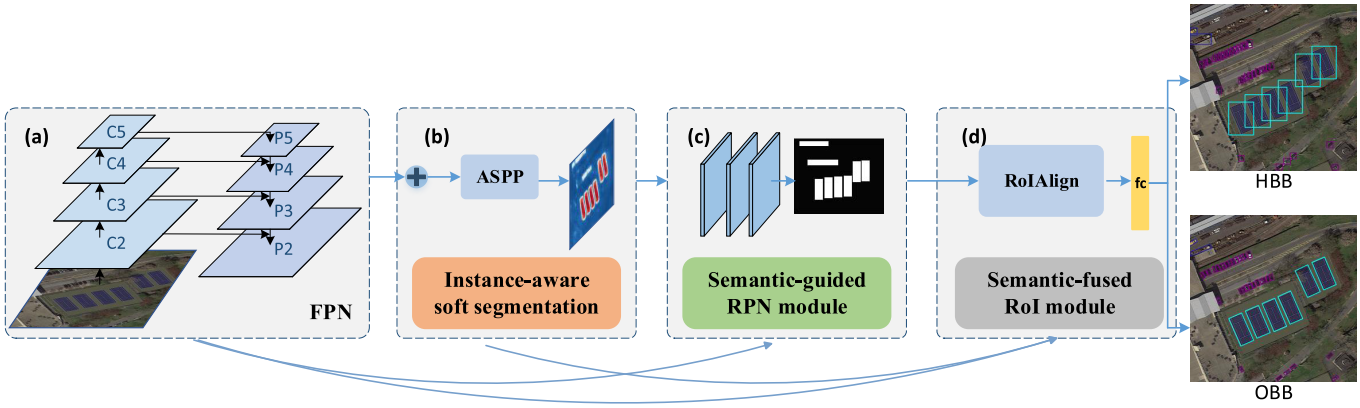


Fig. 1. Illustration of our proposed HSP framework. The HSP consists of four components/modules. (a) FPN-based feature extraction. (b) Instance-aware soft segmentation. (c) Semantic-guided RPN module. (d) S-RoI module. Given a remote sensing image as input, we first adopt a classic top-down and bottom-up structure (e.g., FPN) to extract multiscale feature representation. The instance-aware soft segmentation module is led into the intermediate layer of our HSP, after which we can utilize the predicted instance-aware segmentation results as an auxiliary for better guiding the generation process of region proposals in the semantic-guided RPN module. By propagating hierarchical semantic information (including pixel-level features, multiscale representation from RPN, instance-aware segmentation information, and region proposals) into the RoI module, we can finally address the prediction of HBB and OBB tasks in an end-to-end network architecture.

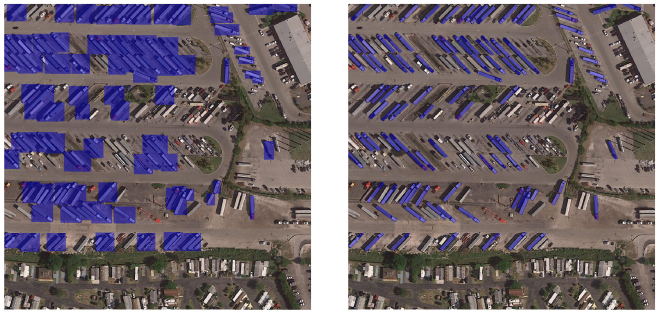
pixel-level features X, the instance-aware semantic information S from the soft-segmentation module, and the predicted region proposals P. The S-RoI module is formulated as

$$F_{\text{fea}} = f_{\text{RoI}}(P, S, R, X). \tag{3}$$

With support from the HSP outlined above, we can finally predict the object category information c, the corresponding HBB $(\text{hbb}_x, \text{hbb}_y, \text{hbb}_w, \text{hbb}_h)$ and OBB $(\text{obb}_x, \text{obb}_y, \text{obb}_w, \text{obb}_h, \text{obb}_\theta)$ in an end-to-end fashion.

B. Instance-Aware Soft Segmentation

Existing object detection methods utilized in remote sensing imagery are easily influenced by more complex situations, including background clutter and the dense arrangement of objects. Moreover, jointly solving detection and segmentation problems has proven to be an effective way of improving the performance of both tasks in [5], [7], and [16]. Drawing inspiration from these excellent works, we introduce the soft segmentation module that aims to obtain this instance-aware semantic information in the HSP detection framework; however, the question of how to obtain the objective/ground-truth of segmentation in the detection task remains an open one. In natural scenes, in order to achieve satisfactory segmentation results, most previous methods (see [5], [6], [7], [48]) take precise pixel-wise annotations as the supervision signal during training. However, there is usually no precise enough mask in the aerial images; moreover, the horizontal box-wise mask is not an effective choice for supervision, due to the high amount of noise in the special bird's-eye-view image. As shown in Fig. 2(a), if we use HBBs to generate the mask, too much useless noise will be included, such as redundant background information or other objects that are adjacent but belong to different object instances. Here, we use the rotation box-wise mask as supervision, since these masks can overlap well with oriented objects rather than introducing disturbance, as illustrated in Fig. 2(b).



(a) (b)

Fig. 2. Two ways of generating a segmentation mask based on two different bounding boxes. Segmentation mask according to the given (a) horizontal boxes and (b) orientation boxes.

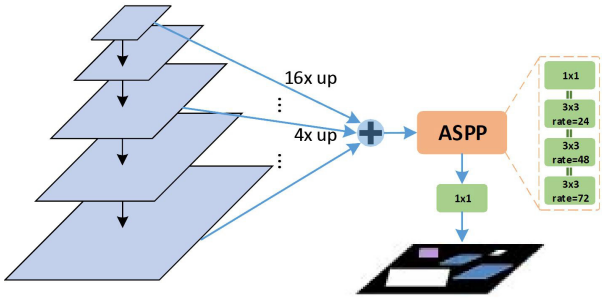


Fig. 3. Overview of instance-aware soft segmentation module, where || refers to the concatenate operation.

In order to estimate the instance-aware segmentation results from remote sensing imagery, we introduce an auxiliary soft segmentation module in the intermediate layer of the HSP network. The detail configuration of the instance-aware soft segmentation module is presented in Fig. 3. Following the feature extraction of FPN, we normalize these extracted multiscale representations into the same spatial scale. As described in HTC [7], the spatial stride of P_3 is 8, such that a large number of the small objects in aerial images (e.g., vehicles and ships) will elicit only a very low or even

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no response. Thus, we choose a larger scale (P_2, with stride 4) to predict the instance-aware semantic segmentation results, which may constitute an appropriate tradeoff between computation and performance. After merging these normalized multiscale representations via the element-wise addition operation, we then utilize an ASPP unit [6] to encode the global semantic information from different receptive fields. As in [6], we use four parallel convolution layers with 256 channels (each followed by ReLU) in the ASPP unit: these comprise one 1×1 convolution layer and three 3×3 convolution layers with different rates (e.g., 24, 48, and 72), where the setting of the atrous rates can influence segmentation and improve the final detection results to some extent. We then concatenate the features from the ASPP unit with a 1×1 convolution layer (with 256 channels), after which we can predict the final instance-aware segmentation results.

C. Semantic-Guided RPN Module

In the original Faster R-CNN [10], [27], RPN is used to generate a series of rough region proposals that cover the objects of interest to varying degrees. These region proposals are then fed into the RoI module to enable more accurate bounding box regression and classification. In FPN [27], RPN is performed separately at each level with shared parameters, where features from the FPN backbone go through one 3×3 convolution layer to produce the head feature, while two parallel 1×1 convolution layers are used to output the bounding box offset and object recognition results. However, these previous RPN processes do not contain any guided semantic information, meaning that it may contain miscellaneous background noise that will have harmful effects on detection. We thus propagate our obtained semantic segmentation results into the RPN module to enable the improved generation of horizontal region proposals. More specifically, we perform the dot product multiplication operation between the segmentation results and multiscale representation of FPN in order to suppress the background information and improve the object-related features simultaneously. As in [10] and [28], the obtained result is then fed into one 3×3 convolution layer to produce the corresponding head feature of RPN, thereby resulting in better region proposal generation.

D. S-RoI Module

In general object detection systems, the RoIAlign [5] or RoIPool [9] operations are used to crop RoI-relevant features from specific convolutional maps. In Fast/Faster R-CNN [9], [10], the RoI feature is fixedly pooled from the last CNN feature map. Due to the features being obtained from different spatial levels in FPN [27], a mapping formula is used to map RoI to the specific pyramid level based on the RoI size. Although this approach takes the size of objects into account, some objects of extreme size are still not well represented. Take a playground of very large size as an example. Although the size-specific level may be appropriate if the target size could be P_5, the feature presentation at the P_5 level is too coarse to enable object location due to excessive down-sample operations. Recently, PANet [49] has proposed the adaptive feature pooling method to fuse pooled features from different

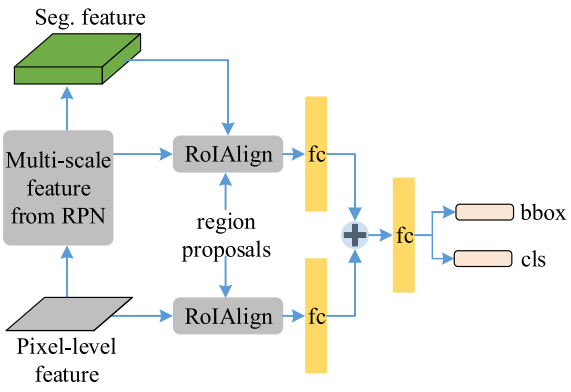


Fig. 4. Overview of the S-RoI module.

levels with element-wise max or sum operations, which can to some extent improve instance segmentation and object detection performance. Here, to boost the discriminative capability of feature representation, we design an S-RoI module to merge hierarchical semantic features together, including multilevel pyramid features from FPN, instance-aware semantic feature from the soft segmentation module, and the pixel-level image information.

The configuration of our S-RoI module is illustrated in Fig. 4. First, we use RoIAlign [5] to crop the features of each horizontal region proposal on the multilevel pyramid features of FPN [27], then perform the same crop operation on the semantic segmentation feature and the normalized pixel-level feature information. We next take several fc layers to fuse these features, which have been pooled from different sources. One part of the pooled features (from the pyramid network and the semantic segmentation feature) can be fed first into one shared fc layer (named fc_1), while another part of the pooled features (from the pixel-level feature) can be sent to another parallel fc layer (named fc_2). Both fc layers, which have the same channel size of 1024, are followed by the ReLU operation. After that, we add up these feature vectors in order to fuse their semantic information, and use one extra fc layer to generate the final proposal-wise feature representation vector. The feature aggregation operation can be represented as follows:

$$F_{\text{fea}} = \sum_{k=2}^{5} \text{fc}_1(\rho(R_k, P)) + \text{fc}_1(\rho(S, P)) + \text{fc}_2(\rho(X_{\text{norm}}, P)) \tag{4}$$

where F_{fea} refers to the final fused features of the RoI module, X_{norm} is the normalized pixel-level image feature, R and S are the kth level pyramid feature and semantic segmentation feature of the whole image, respectively. P denotes the set of horizontal region proposals outputted by RPN, whereas ρ stands for the RoIAlign operation. Finally, as in [10], we employ the obtained discriminative feature information F_{fea} to predict the categories of objects, and the corresponding HBBs and OBBs in the remote sensing imagery.

E. HSP Training

In order to perform object classification, the regression of HBBs and OBBs, and the instance-aware segmentation

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simultaneously in a unified network, we design a multitask objective function as follows:

$$\begin{aligned} L_{\mathrm{HSP}} &= \lambda_1 \frac{1}{N_{\mathrm{cls}}} \sum_i L_{\mathrm{cls}}\left(p_i, p_i^*\right) \\ &+ \lambda_2 \frac{1}{N_{\mathrm{reg}_{hbb}}} \sum_i p_i^* L_{\mathrm{reg}_{hbb}}\left(\mathbf{h}_i, \mathbf{h}_i^*\right) \\ &+ \lambda_3 \frac{1}{N_{\mathrm{reg}_{obb}}} \sum_i p_i^* L_{\mathrm{reg}_{obb}}\left(\mathbf{o}_i, \mathbf{o}_i^*\right) \\ &+ \lambda_4 \frac{1}{N_{\mathrm{seg}}} \sum_{li} \sum_{lj} L_{\mathrm{seg}}\left(s_{li,lj}, s_{li,lj}^*\right). \end{aligned} \tag{5}$$

Here, i is the index of bounding box, p_i represents the predicted probability of the ith box being an object, and the groundtruth category p_i^* is 0 if this box is negative and 1 for positive ones. $\mathbf{o}_i^*$ and $\mathbf{h}_i^*$ are the coordinates of the groundtruth OBBs and HBBs, whereas $\mathbf{o}_i$ and $\mathbf{h}_i$ represent the parameterized coordinate vectors of the estimated OBB and HBB, respectively. $s_{li,lj}^*$ and $s_{li,lj}$ refer to the groundtruth mask label and the predicted mask score at each position (li, lj). The classification loss term L_{cls} in (5) is a cross entropy loss function across all categories. For the location regression objective functions of the OBB and HBB tasks, we adopt $L_{\mathrm{reg_{obb}}}(\mathbf{o}_i, \mathbf{o}_i^*) = R(\mathbf{o}_i - \mathbf{o}_i^*)$ and $L_{\mathrm{reg_{hbb}}}(\mathbf{h}_i, \mathbf{h}_i^*) = R(\mathbf{h}_i - \mathbf{h}_i^*)$, where R is a smooth L1 loss function defined in [9]. The terms $p_i^* L_{\mathrm{reg_{hbb}}}(\mathbf{h}_i, \mathbf{h}_i^*)$ and $p_i^* L_{\mathrm{reg_{obb}}}(\mathbf{o}_i, \mathbf{o}_i^*)$ mean the HBB and OBB regression objective function are activated only for these positive boxes ($p_i^* = 1$) and are disabled otherwise ($p_i^* = 0$). For the term L_{seg}, we utilize the focal loss [37] as the semantic segmentation objective function. The four terms N_{cls}, $N_{\mathrm{reg_{hbb}}}$, $N_{\mathrm{reg_{obb}}}$, and N_{seg} are the normalization factors to reduce the influences of different scales/sizes when calculating losses. N_{cls}, $N_{\mathrm{reg_{hbb}}}$, and $N_{\mathrm{reg_{obb}}}$ are the number of sampled boxes for loss calculation during the training process, and N_{seg} refer to the size of predicted mask [i.e., $N_{\mathrm{seg}} = (W/4)*(H/4)$] in our article. Note that we have $N_{\mathrm{cls}} = N_{\mathrm{reg_{hbb}}} = N_{\mathrm{reg_{obb}}}$ since the common sampled boxes. The hyper-parameters λ_1, λ_2, λ_3, and λ_4 are the balance factors of the four loss terms. For simplicity, we set $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 1$ for all experiments in this article.

For both HBB and OBB tasks, we regress the central point (x, y), width and height (w, h), and especially the extra angle θ for the OBB task. The definition of θ is the same as in OpenCV and is converted into radians with a range of $[-\pi/2, 0)$. The illustration of predicted, anchor, and groundtruth boxes for both HBB and OBB tasks can be seen in Fig. 5. In the regression process of the HBB task, we employ the parameterizations of the four coordinates according to the following method:

$$\begin{aligned} \mathrm{hbb}_x &= (x - x_a)/w_a, \quad \mathrm{hbb}_y = (y - y_a)/h_a \\ \mathrm{hbb}_w &= \log(w/w_a), \quad \mathrm{hbb}_h = \log(h/h_a) \\ \mathrm{hbb}_x^* &= (x^* - x_a)/w_a, \quad \mathrm{hbb}_y^* = (y^* - y_a)/h_a \\ \mathrm{hbb}_w^* &= \log(w^*/w_a), \quad \mathrm{hbb}_h^* = \log(h^*/h_a) \end{aligned} \tag{6}$$

where x, x_{a}, and x^* are the variables for the predicted box, anchor box, and ground-truth box of the HBB task, respectively, and similarly for y, w, h. For the predicted box and groundtruth box of HBB task, the parameterizations of four coordinates are $\mathbf{h} = [\mathrm{hbb}_x, \mathrm{hbb}_y, \mathrm{hbb}_w, \mathrm{hbb}_h]$ and $\mathbf{h}^* = [\mathrm{hbb}_x^*, \mathrm{hbb}_y^*, \mathrm{hbb}_w^*, \mathrm{hbb}_h^*]$. For the OBB regression, we also adopt the same parameterization operation for the four coordinates; however, an oriented parameter will be processed as follows:

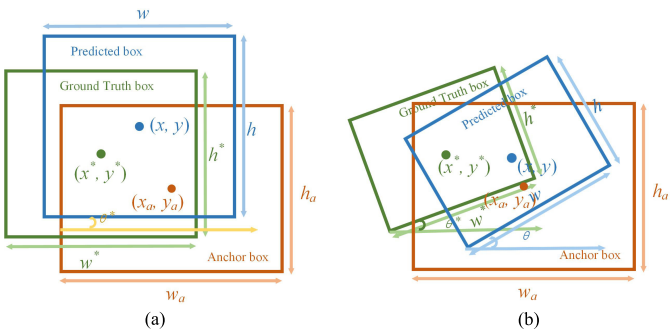


Fig. 5. Illustration of the predicted, anchor, and groundtruth boxes for (a) HBB and (b) OBB tasks, respectively.

$$\mathrm{obb}_\theta = \theta - \theta_a, \quad \mathrm{obb}_\theta^* = \theta^* - \theta_a \tag{7}$$

where θ, θ_{a}, and θ^* are the oriented variables for the predicted box, anchor box, and groundtruth box of the OBB task. Here in order to reduce the computational complexity of proposal generation for the OBB task, we regression these rotated bounding boxes from horizontal region proposals, i.e., θ_{a} is always equal to $-90°$ in our article. Similarly, the parametric forms of the predicted box and ground-truth box are $\mathbf{o} = [\mathrm{obb}_x, \mathrm{obb}_y, \mathrm{obb}_w, \mathrm{obb}_h, \mathrm{obb}_\theta]$ and $\mathbf{o}^* = [\mathrm{obb}_x^*, \mathrm{obb}_y^*, \mathrm{obb}_w^*, \mathrm{obb}_h^*, \mathrm{obb}_\theta^*]$, respectively.

IV. Experiments

A. Experimental Setup

1) Data Sets: To evaluate our proposed HSP, we conduct comprehensive experiments on three public aerial data sets: namely DOTA [34], UCAS_AOD [25], and NWPU VHR-10 [24]. Although the DOTA and UCAS_AOD data sets contain both OBB and HBB tasks, NWPU VHR-10 data set contains the HBB task only. The evaluation protocol for all three data sets follows the PASCAL VOC benchmark [50], using mAP as the primary metric. A predicted bounding box is assigned to match a ground truth box if its IoU is larger than 0.5. For the OBB task, the intersection over the union area of two polygons is adopted to calculate the IoU.

DOTA [34] is the largest data set for object detection in aerial images at present. It contains 2806 aerial images with sizes ranging from 800 × 800 to 4000 × 4000 pixels, and comprises objects belonging to 15 categories with 188 282 instances in total. These objects have arbitrary scales, orientations and shapes, each of which is annotated on an attached quadrilateral. Categories are as follows: plane, baseball diamond (BD), bridge, ground track field (GTF), small vehicle (SV), large vehicle (LV), ship, tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA),

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TABLE I
QUANTITATIVE COMPARISONS BETWEEN THE BASELINES AND OUR HSP METHOD FOR OBB TASK ON DOTA TEST SET

Method	mAP(%)	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC
SSD [11]	10.59	39.83	9.09	0.64	13.18	0.26	0.39	1.11	16.24	27.57	9.23	27.16	9.09	3.03	1.05	1.01
YOLOv2 [13]	21.39	39.57	20.29	36.58	23.42	8.85	2.09	4.82	44.34	38.25	34.65	16.02	37.62	47.23	25.50	7.45
R-FCN [52]	26.79	37.80	38.21	3.64	37.26	6.74	2.60	5.59	22.85	46.93	66.04	33.37	47.15	10.60	25.19	17.96
FR-H [10]	36.29	47.16	61.00	9.80	51.74	14.87	12.80	6.88	56.26	59.97	57.32	47.83	48.70	8.23	37.25	23.05
FR-O [10]	52.93	79.09	69.12	17.17	63.49	34.20	37.16	36.20	89.19	69.60	58.96	49.40	52.52	46.69	44.80	46.30
ICN [20]	68.16	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23
RoI-Transformer [23]	69.56	88.64	78.52	43.44	75.92	68.81	**73.68**	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
R^2CNN++ [33]	71.16	89.66	81.22	45.50	75.10	68.27	60.17	66.83	**90.90**	80.69	**86.15**	64.05	63.48	65.34	68.01	62.05
Ours	**78.01**	**90.39**	**86.23**	**56.12**	**80.59**	**77.52**	73.26	**83.78**	90.80	**87.19**	85.67	**69.08**	**72.02**	**76.98**	**72.50**	**67.96**

TABLE II
QUANTITATIVE COMPARISONS BETWEEN THE BASELINES AND OUR HSP METHOD FOR HBB TASK ON DOTA TEST SET

Method	mAP(%)	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC
SSD [11]	10.94	44.74	11.21	6.22	6.91	2.00	10.24	11.34	15.59	12.56	17.94	14.73	4.55	4.55	0.53	1.01
YOLOv2 [13]	39.20	76.90	33.87	22.73	34.88	38.73	32.02	52.37	61.65	48.54	33.91	29.27	36.83	36.44	38.26	11.61
R-FCN [52]	47.24	79.33	44.26	36.58	53.53	39.38	34.15	47.29	45.66	47.74	65.84	37.92	44.23	47.23	50.64	34.90
FR-H [10]	60.46	80.32	77.55	32.86	68.13	53.66	52.49	50.04	90.41	75.05	59.59	57.00	49.81	61.69	56.46	41.85
ORConv [40]	70.10	89.30	73.80	47.70	52.80	73.90	73.90	87.20	90.80	79.30	81.10	47.50	56.10	73.50	70.90	53.60
ICN [20]	72.45	89.97	77.71	53.38	73.26	73.46	65.02	78.22	90.79	79.05	84.81	57.20	62.11	73.45	70.22	58.08
IAD R-CNN [24]	72.72	88.62	80.22	53.18	66.97	76.30	72.59	84.07	90.66	80.95	76.24	57.12	66.65	74.08	66.36	56.85
R^2CNN++ [33]	75.35	90.18	81.88	55.30	73.29	72.09	77.65	78.06	**90.91**	82.44	86.39	64.53	63.45	75.77	78.21	60.11
DM-FPN [34]	79.30	88.70	85.10	62.10	78.90	**78.20**	**83.30**	**87.30**	90.80	84.80	**87.10**	68.30	69.30	83.30	78.20	64.00
Ours	**80.42**	**90.42**	**86.91**	**62.57**	**79.96**	78.13	81.86	85.27	90.80	**87.30**	85.94	**69.96**	**72.11**	**84.13**	**80.99**	**69.88**

harbor, swimming pool (SP), and helicopter (HC). The DOTA data set is split into training (1/2), validation (1/6), and test sets (1/3). UCAS_AOD [25] consists of 1510 aerial images of approximately 1000 × 1000 pixels, of which 1000 images contain planes (with 7482 instances) and 510 images contain cars (with 7114 instances). Following [19] and [34], we randomly select 1110 images for training and 400 images for testing: the selected images contain 708 images of planes and 402 images of cars for training, as well as 292 plane images and 108 car images for testing.

NWPU VHR-10 [24] is a publicly available geospatial object detection data set with ten categories, similar to the DOTA data set; however, the groundtruth location information of all objects is indicated only with HBBs. This data set contains a total of 800 (650 positive and 150 negative) very high-resolution (VHR) remote sensing images that have been cropped from Google Earth and the Vaihingen data set and then manually annotated by experts. To facilitate a fair comparison with baseline methods [24], [41], [46], we divide the positive data set into 20% for training, 20% for validation and 60% for testing.

2) Implementation Details: For the DOTA data set, we split the images into subimages (1024 × 1024 pixels in size) with 512 pixel overlap via the official development kit, and we also use the multiscale technique in both training and testing processes. More specifically, we first resize the original images by 1.5× and 0.5× before splitting, then take all subimages (along with the original split ones) as the final training and testing sets. As for the UCAS_AOD and NWPU VHR-10 data sets, we resize images with long edges to 1024 pixels and those with short edges to 832 pixels. For baseline, we simply extend FPN-based Faster R-CNN [27] via an additional rotation branch such as R^2CNN [30]. During training, we use the pretrained ResNet-101 [2] to initialize our backbone. For the DOTA and UCAS_AOD data sets, we train the model for 12 epochs with batch size 4 on 4 T P40 GPUs (effective minibatch is 16) with a learning rate of 0.02, and decrease the learning rate by 10 after 8 and 11 epochs. The weight decay is set to 0.0001 and the momentum is 0.9. For the NWPU VHR-10 data set, we only utilize training set of 120 images and evaluate our method on test set containing 410 images; and we also assess the effectiveness of our method only for the HBB task. For this data set, unlike the other two data sets used for these experiments, the supervised information of semantic segmentation and the final loss function comes from HBBs in the training process. Since the number of images is too small, we train the model using far more epochs and a lower learning rate compared to the other two data sets. We train it for a total of 120 epochs with batch size 2 on a single GPU, with a learning rate of 0.0005, which is decreased by 10 after 80 and 110 epochs. Throughout the whole training process, we also implement data augmentation including random flip and rotation.

B. Comparisons With State-of-the-Art

We compare our proposed HSP with several state-of-the-art approaches [22], [32], [33], [41], [45] on three data sets.

1) DOTA Data Set [34]: Tables I and II show the performance of our HSP and comparisons with several state-of-the-art methods on both OBB and HBB tasks, respectively. The corresponding visualization results for both OBB and HBB object detection tasks on the DOTA test set are presented in Fig. 6. For the OBB task, our HSP significantly outperforms these baselines, achieving improvements of 6.85% over R^2CNN++ [32], 8.45% over RoI-Transformer [22], 9.85% over ICN [19] and 25.08% over FR-O [10] in terms

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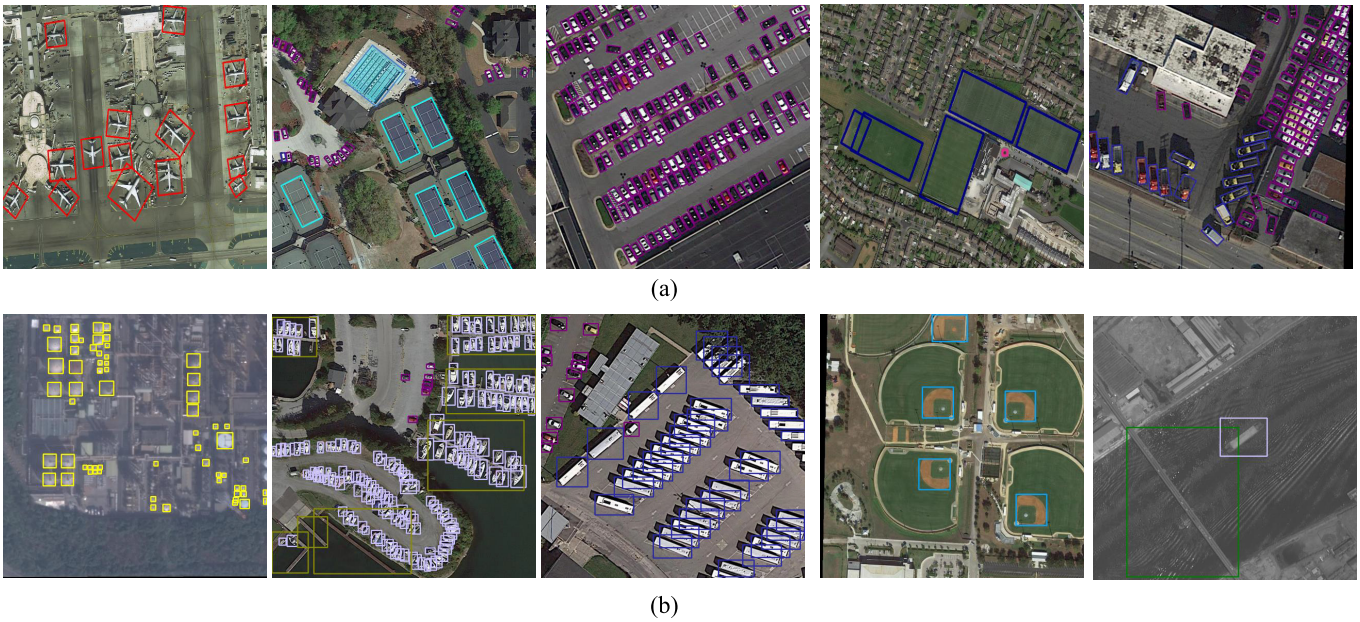


Fig. 6. Examples of our detection results for both OBB and HBB tasks on DOTA test set. (a) Visualization of OBB predictions on the DOTA test set. (b) Visualization of HBB predictions on the DOTA test set.

TABLE III
RESULT COMPARISONS BETWEEN THE BASELINES AND OUR HSP METHOD FOR HBB TASK ON NWPU VHR-10 TEST SET

Method	mAP(%)	Plane	Ship	ST	BD	TC	BC	GTF	Harbor	Bridge	Vehicle
RICNN [25]	72.63	88.35	77.34	85.27	88.12	40.83	58.45	87.63	68.60	61.51	71.10
R-P-Faster R-CNN [41]	76.50	90.40	75.00	44.40	89.90	79.00	77.60	87.70	79.10	68.20	73.20
Deformable R-FCN [45]	79.10	87.30	81.40	63.60	90.40	81.60	74.10	90.30	75.30	71.40	75.50
Deformable Faster RCNN [46]	84.40	90.70	87.10	70.50	89.50	89.30	87.30	97.20	73.50	69.90	88.80
RDAS512 [47]	89.50	99.60	85.50	89.00	95.00	89.60	**94.80**	95.30	82.60	77.20	86.50
Multi-Scale CNN [42]	89.60	99.30	92.00	83.20	97.20	**90.80**	92.60	98.10	85.10	71.90	85.90
R^2CNN++ [33]	91.75	**100**	89.41	**97.22**	97.00	83.15	87.54	**99.17**	**99.40**	74.51	**90.10**
Ours	**93.38**	99.79	**92.45**	96.96	**98.55**	90.37	91.48	99.04	88.90	**87.14**	89.07

of mAP score. When compared with R^2CNN++ [32], our method for the OBB task also achieves a very large gain in most categories: e.g., 56.12% versus 45.50% for bridge, 77.52% versus 68.27% for SV, 87.19% versus 80.69% for BC, 72.02% versus 63.48% for RA, and 76.98% versus 65.34% for harbor. This indicates that our HSP performs very well on detecting these dense and irregularly oriented objects. Meanwhile, we can achieve 80.42% mAP for the HBB task on DOTA data set, which is significantly better than the baselines, representing improvements of 1.12%, 5.07%, 7.7%, and 7.79% over DM-FPN [33], R^2CNN++ [32], IAD R-CNN [23] and ICN [19]. Moreover, for both the OBB and HBB tasks, we achieved better performance in discriminating between similar categories, such as GTF and SBF or plane and HC, than previous methods [10], [32], [33] These improvements may be primarily due to the fact that, by simultaneously considering HSPs, the label discriminations can be well captured during the category prediction process, especially for categories with similar appearance characteristics.

2) NWPU VHR-10 [24] and UCAS_AOD [25] Data Sets: We first report the results of comparisons between the baselines and our HSP method for the HBB task on NWPU VHR-10 [24]. As can be seen from Table III, we obtain 93.38% mAP and achieve the best performance in several object categories: e.g., 92.45% for ship, 98.55% for BD, and 87.14% for bridge, respectively. Comparisons with previous methods [32], [41], [44]–[46] demonstrate that our proposed HSP also substantially outperforms them, achieving improvements of 1.63% over R^2CNN++ [32], 3.78% over multiscale CNN [41], 8.98% over deformable faster RCNN [45], and 20.75% over RICNN [24] in terms of mAP score. This demonstrates that our HSP method performs very well even on a single HBB task in the remote sensing imagery. Furthermore, we also report the detection results of our HSP and the existing methods [19], [34] on the UCAS_AOD data set [25] with only two categories. As shown in Table IV, we achieve 95.99% and 97.00% performance in terms of mAP score for OBB and HBB tasks. Since the results of each category are not reported in the ICN [19], we have also shown the overall performance here; note that ICN [19] utilizes the joint image cascade and feature pyramid, which results in a fairly large computational burden for both the training and testing processes compared to our HSP method. Furthermore, our proposed HSP method also achieves better performance than ICN [19] for the OBB task. For the HBB task, we significantly outperform the baseline Xia *et al.* [34], e.g., 97.00% versus

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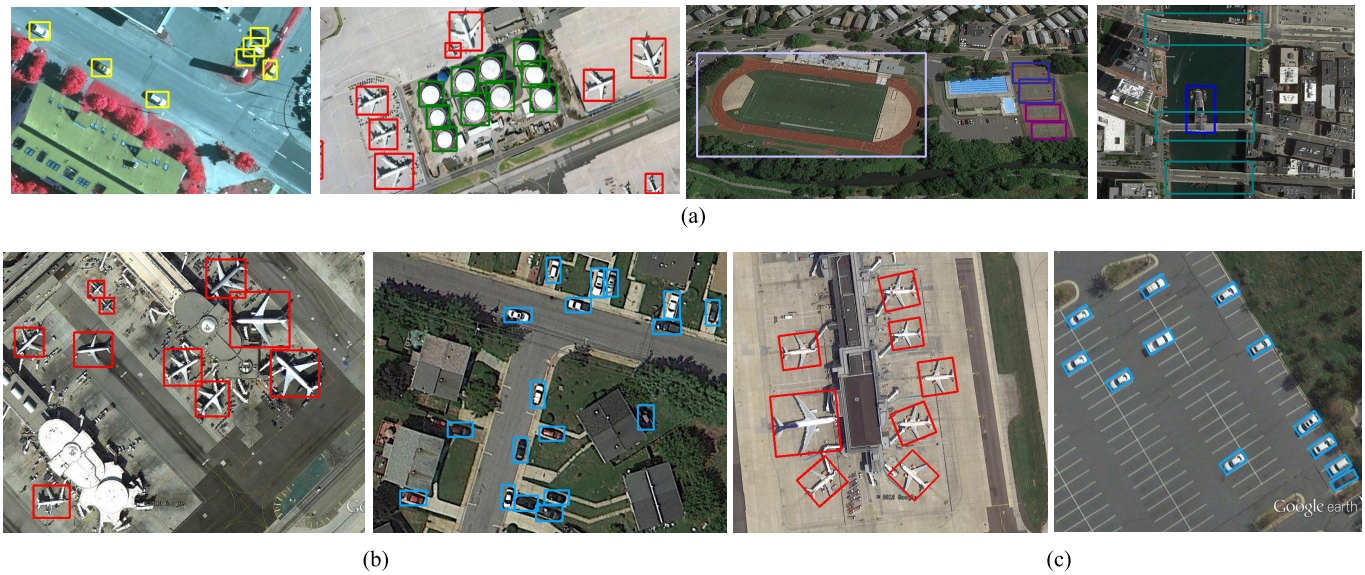


Fig. 7. Visualization of our detection results on UCAS_AOD and NWPU VHR-10 data sets. (a) Samples of HBB predictions on NWPU VHR-10 test set. (b) Samples of HBB predictions on UCAS_AOD test set. (c) Samples of OBB predictions on UCAS_AOD test set.

TABLE IV
RESULT COMPARISONS ON UCAS-AOD TEST SET

Task	Method	mAP(%)	Plane	Car
OBB	ICN [20]	95.67	–	–
	Ours	**95.99**	**98.36**	**93.61**
HBB	Xia *et al.* [35]	89.41	90.66	88.17
	Ours	**97.00**	**98.77**	**95.22**

TABLE V
COMPARISON WITH DIFFERENT COMPONENTS ON DOTA TEST SET

Baseline	S-RPN	S-RoI	mAP@OBB(%)	mAP@HBB(%)
✓	–	–	73.94	76.57
✓	✓		76.10	77.67
✓		✓	77.22	78.52
✓	✓	✓	**78.01**	**80.42**

89.41% for the score of HBB task, 98.77% versus 90.66% for the plane category, and 95.22% versus 88.17% for the car category. Some visualization results of our HSP method on the UCAS_AOD and NWPU VHR-10 test sets can be found in Fig. 7. These above experimental results indicate that our proposed HSP method can better predict the object category information and the corresponding locations by improving the predictive capability of the network.

C. Ablation Study

1) Componentwise Analysis: As presented in Table V, we first evaluate the different network settings with different components/modules on the DOTA test set, including the instance-aware soft-segmentation module, semantic-guided RPN and S-RoI module. In more detail, the baseline network refers to an extended Faster R-CNN, such as R^2CNN [30] that does not use any semantic information.

The baseline with semantic-guided RPN module (i.e., baseline + S-RPN) takes the output produced from the soft-segmentation branch to re-weight the original features of the RPN module. The baseline with S-RoI module (i.e., baseline + S-RoI) simultaneously considers pooled features from FPN, semantic-aware features, and the original image in the RoI module but does not use any semantic-guided information in the RPN. When using only semantic information in the RPN module, the performance of "baseline + S-RPN" can be improved by 2.16% and 1.1% for the OBB and HBB tasks, compared with the baseline network. The mAP scores of "baseline + S-RoI" are also better than the baseline network, e.g., 77.22% versus 73.94% for the OBB task, 78.52% versus 76.57% for the HBB task. After employing both the semantic-guided RPN and S-RoI (namely, our HSP network), we can further improve the detection performance with 78.01% and 80.42% for OBB and HBB tasks, respectively. This demonstrates that the semantic information from the soft-segmentation module can be beneficial to the proposal generation in RPN and feature fusion in the RoI stage and finally improves the network performance when predicting the object category and corresponding horizontal and oriented locations.

In order to accelerate the below ablation studies, we train/optimize the proposed HSP network on the DOTA training set, and then analyze it on the DOTA validation set.

2) Semantic Segmentation Module: We further analyze the predictive capability of the semantic segmentation module under different network settings, and also report the corresponding detection performance of our HSP on the DOTA validation set. Here, mIoU is used to measure the difference in score between the predicted segmentation result and the ground-truth obtained from the OBB. Qualitative comparisons of the segmentation and detection results are reported in Table VI.

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TABLE VI

COMPARISONS WITH VARIOUS SETTINGS OF THE SEMANTIC SEGMENTATION MODULE (S) ON DOTA VALIDATION SET

Method	Setting	mAP@OBB(%)	mAP@HBB(%)	mIoU(%)
Baseline	–	76.23	78.74	–
HSP + S w/ Conv×4	–	77.23	79.89	53.34
HSP + S w/ ASPP [6]	(6, 12, 18)	77.65	79.98	54.81
	(12, 24, 36)	77.83	80.03	55.02
	(24, 48, 72)	**78.10**	**80.50**	**55.27**
	(48, 96, 144)	77.47	80.38	54.55
	(24, 48, 72) + IP	77.27	79.17	53.62
HSP + S w/ PSP [49]	(1, 2, 3, 6)	77.77	79.91	54.42
	(1, 8, 24, 56)	77.09	79.97	52.68
HSP + S w/ GCN [53]	k = 15	77.51	79.94	53.02
	k = 31	77.27	80.09	53.48

TABLE VII

ABLATION STUDY OF VARIOUS SETTINGS IN THE S-ROI MODULE ON DOTA VALIDATION SET

s-scale	m-scale	pixel	semantic	OBB	HBB
✓	-	-		76.23	78.74
-	✓	-		77.28	79.74
-	✓	✓		77.45	80.22
-	✓	✓	✓	**78.10**	**80.50**

TABLE VIII

COMPUTATIONAL REQUIREMENT COMPARISONS ON DOTA VALIDATION SET

Method	Image	Memory (GB)	Training (s/iter)	Testing (FPS)
Faster R-CNN [28]	1024×1024	9.50	0.29	4.5
Baseline network	1024×1024	11.48	0.39	3.7
Our method	1024×1024	12.80	0.45	3.5

The baseline algorithm refers to the detection network that operates without using any instance-level semantic information in all modules. "Conv × 4" means that we use four consecutive 3 × 3 convolution layers to replace the ASPP unit in the semantic segmentation module. In the bottom four rows, we report the performance with different atrous rates in the ASPP unit of the semantic segmentation module, including (6, 12, 18), (12, 24, 36), (24, 48, 72), and (48, 96, 144). From the experimental results, we can observe that HSP boosts the detection performance compared to the baseline with the help of semantic segmentation. When we use the ASPP unit by increasing the atrous rates of the ASPP unit, the predicted results can be further improved. However, if we increase the atrous rates to (48, 96, 144), the detection mAP and segmentation mIoU will drop slightly, thus we use (24, 48, 72) as the atrous setting in all our experiments. The reason for this may be that some small objects, such as SVs, elicit a very low response when the atrous rates are increased, meaning that the overall performance drops to some extent. Next, we also evaluate the IP mentioned in DeepLabv3 [6] on the DOTA data set. However, when we add the IP to the ASPP module whose setting achieved the best results, the detection results on both OBB and HBB tasks are declined to a certain extent, and the segmentation results with respect to mIoU decrease too. This phenomenon can be attributed to the characteristic of remote sensing imagery. As small-scale objects account for the major quantity in the DOTA data set, such as SV, LV, and ship, it will result in a more vast background area than the foreground area, which is much different from those natural images like COCO [5]. But actually the IP operation calculates the average number of all the pixels of each channel. If we add it to the segmentation module, too much noise will be introduced to lead to poor results.

In addition, we conduct more experiments about semantic propagation with various settings, such as PSPNet [48] and GCN [52], which are listed in the remaining rows of Table VI. Since the PSP module contains the same IP operation, its results are basically the same as ASPP with IP. Accordingly, the mIoU becomes better when pooling bin sizes are set to (1, 2, 3, 6), but the detection results become poor. Experiments on the GCN [52] show that a larger receptive field (larger kernel size) is important, which is consistent with the conclusion of the ablation study of atrous rates in ASPP. However, GCN only uses one kernel size ($k * k$ as factorized $k * 1$ and $1 * k$) that will lead to the loss of important information compared to ASPP. Although when $k = 31$ brings a bit gain compared to $k = 15$ on the HBB and segmentation tasks, the parameters have increased nearly four times.

3) S-RoI Module: As presented in Table VII, we report the predicted detection results on the DOTA validation set. Here, the S-RoI module can be set with different feature propagations, including single-scale FPN feature (s-scale), multiscale FPN feature (m-scale), pixel-level representation (pixel), and segmentation-guided feature (semantic).

When employing only the single-scale FPN feature in the RoI module, we can obtain 76.23% and 78.74% for OBB and HBB tasks, respectively. The detection results with a multiscale FPN feature can be improved by 1.03% and 1.0% for OBB and HBB tasks, compared with the single-scale FPN feature. After further increasing the pixel-level representation and semantic-guided features, the detection results for OBB and HBB tasks are 78.10% and 80.50% in terms of mAP score, representing improvements of 1.87% and 1.76% relative to the baseline (i.e., only with the single-scale FPN feature in the RoI module). This demonstrates that the detection results on both tasks exhibit gradual improvement as more semantic features are utilized in the RoI stage.

4) Computational Cost: We finally provide the computational complexity analysis between our HSP method and the baseline network, as can be seen in Table VIII. For the baseline network, we extend FPN-based Faster R-CNN [27] via an additional rotation branch such as R^2CNN [30], and note that we do not use other components mentioned in R^2CNN, like multiple pooled sizes. As a comparison, we also trained Faster R-CNN [10] which based on FPN [27] on COCO data set [53] with the same image size (1024 × 1024) as used in our method. In Table VIII, we give the memory requirement, training and testing speeds on a single GPU of Tesla P40. Our proposed HSP spends a little more memory and time than the baseline network, e.g., 11.48 versus 12.80 GB for the memory consumption, 3.70 versus 3.50 FPS in the testing process. The extra time and memory mainly come

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from instance-aware segmentation predictions and the fusion of semantic information, but they just account for a small ratio in the entire process. But our proposed method is greatly superior to the baseline network on DOTA validation set as reported in the manuscript. Thus our HSP method is also efficient in practice.

V. Conclusion

In this article, we propose a novel HSP framework, designed to solve the object detection problem in remote sensing imagery, which can better utilize and propagate multiscale representation and instance-aware semantic information in a unified network. For each input sample, the general FPN is used to extract multiscale feature representations to capture the local detailed information and global semantic structure. We then introduce an instance-aware soft segmentation module in the middle layer of the HSP network; its predictions can be propagated to support the generation of region proposals in the RPN module. Moreover, to further consider hierarchical semantic information, we simultaneously fuse these propagated features in the RoI module, including multiscale representation, soft segmentation results, the generated region proposals, and the pixel-level image features. Finally, we produce the category information and the corresponding HBBs and OBBs in an end-to-end HSP framework. Extensive experimental results have clearly demonstrated the effectiveness of the proposed HSP in solving the object detection problem in remote sensing imagery. In the future, we will further extend our HSP architecture to address vision understanding tasks, especially in aerial scenes.

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